SUMIT KUMAR

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SUMMARY

Final-year B.Tech (AI & ML) student at NIT Kurukshetra focused on building scalable backend and distributed systems, with hands-on experience building LLM agents and Advanced RAG pipelines using lanchain and langraph.

EDUCATION

B.Tech, Artificial Intelligence and Machine Learning – National Institute of Technology, Kurukshetra *2023 – Expected 2027*
CGPA: 8.13
Senior Secondary (CBSE) – Aggarwal Public School *2023 – 91%*

EXPERIENCE

Crushaders Tech Solutions, LLP, Bhubaneswar *Remote*
Research Intern – Automation for Social Media Engagement & Reputation Management January 2026 – March 2026

- **Agentic Workflow Automation:** Built an agentic, API-driven scheduling and publishing pipeline in Python across Facebook, Instagram, and X, automating end-to-end content distribution and reducing manual effort.
- **LLM-Powered Response Agent:** Designed LLM response agents for comments, messages, and reviews, using intent detection and structured outputs to generate context-aware replies.
- **AI Output Validation & Guardrails:** Added output validation and guardrail checks to a review-request and spam/fake-review filtering pipeline, improving the reliability of customer-feedback data.

PROJECTS

Scalable URL Shortener & Analytics Platform *React, TypeScript, Node.js, Express.js, MongoDB, Redis, Docker, Nginx, k6*
GitHub

- Built a full-stack URL platform with JWT-secured REST APIs, Base62 short codes, custom aliases, link expiry, and click analytics, backed by optimized MongoDB indexing.
- Ensured high availability with a MongoDB Replica Set, and used a Redis Cluster for cache-aside caching and rate limiting; a single instance sustained 1,618 RPS at 31.9 ms p95 and 48.6 ms p99 under k6 load tests.
- Scaled horizontally with Docker and Nginx from 1 to 6 backend instances, raising throughput from 1,140 to 5,640 RPS (4.9x, 82% scaling efficiency); validated with 37 automated tests across 6 suites.

ResearchPilot – Agentic RAG Research Assistant *Python, LangGraph, LangChain, Qdrant, Streamlit, OpenAI, DeepEval*
Live Demo | GitHub

- Designed a multi-agent system in LangGraph with an LLM router coordinating retrieval, claim-verification, and direct-answer agents, using bounded retries to prevent runaway execution loops.
- Built a RAG pipeline ingesting PDF, TXT, Markdown, URL, and ArXiv sources into Qdrant with cached embeddings, plus web/ArXiv retrieval to ground claim verification.
- Built a DeepEval pipeline over 200+ benchmark queries, reaching 0.91 faithfulness and 0.93 answer relevancy, to serve as quality gates for output reliability.

Microservices-Driven Real-Time Chat System *WebSocket, REST, Redis, Kafka, MongoDB, PostgreSQL, S3, APNs/FCM, Docker, AWS*
Live Demo | GitHub

- Architected a distributed chat system *designed to target* 500M DAU and ∼700K messages/sec peak, with separate WebSocket and REST paths and stateless, horizontally scalable services.
- Implemented a store-first, deliver-second pipeline using Snowflake IDs, per-conversation sequence numbers, and `client_msg_id` idempotency, with Kafka and APNs/FCM push, supporting ordered delivery, offline sync, presence, and read receipts.
- Designed a polyglot data layer: MongoDB for messages, PostgreSQL for users and groups, Redis for TTL-based presence, and S3 for media messaging.

CostMind – Cost-Aware Conversational Agent *Python, LangGraph, LiteLLM, Qdrant, Redis, FastAPI*
GitHub

- Built a memory-enabled agent with LangGraph, FastAPI, Qdrant, and Redis, combining short-term buffers with per-user long-term vector memory ranked by similarity, recency, and importance.
- Implemented cost-aware LLM routing with LiteLLM: query-complexity classification, confidence-based escalation, budgets, provider fallback, and privacy-scoped semantic caching, avoiding 17.8% of LLM calls (57/320).
- Benchmarked 320 queries against an always-GPT-4o baseline: cut LLM cost by 83.2% ($1.27 → $0.21) and prompt tokens by 63.4%, with a 4.7/5 LLM-judged quality score on 60 queries.

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript, SQL
Backend: Node.js, Express.js, FastAPI, React, REST APIs, WebSocket, Microservices, JWT Authentication
System Design: Scalability, High Availability, Fault Tolerance, Load Balancing, Caching, Message Queues (Kafka)
Databases: MongoDB, PostgreSQL, Redis, Qdrant
Cloud & Tools: AWS, Docker, Nginx, Git, Postman, Automated Testing, k6
AI: LLMs, RAG, LangChain, LangGraph, Agentic AI
CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

POSITIONS OF RESPONSIBILITY & ACHIEVEMENTS

Outstanding Mentee – Foundation For Excellence (FFE) Scholarship Program *FY 2025–26*
Volunteer Manager – Anant, The Mathematical Society, NIT Kurukshetra